QUBO Formulation: Legal Document Extractive Summarization

Legal Document Extractive Summarization

Problem Statement. Given a legal document partitioned into N candidate sentences, the goal is to construct an extractive summary of exactly K sentences. Each sentence $i \in \{0, \ldots, N-1\}$ is assigned an importance score $w_i \geq 0$. Additionally, each pair of distinct sentences (i, j) with $0 \leq i < j \leq N-1$ is assigned a redundancy penalty $p_{ij} \geq 0$ reflecting the degree to which they cover overlapping legal points. The objective is to select a subset of exactly K sentences that maximizes the total importance score while minimizing the total redundancy among the selected sentences.

Decision Variables. For each candidate sentence $i \in \{0, \ldots, N-1\}$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i equals 1 if sentence i is included in the summary, and 0 otherwise.

QUBO Derivation. **Step 1.** The original maximization objective is converted to a minimization problem by negating the objective function:

$$\min_{x \in \{0,1\}^N} \left(-\sum_{i=0}^{N-1} w_i x_i + \sum_{0 \leq i < j \leq N-1} p_{ij} x_i x_j \right). \tag{1}$$

Step 2. The exactly-K selection constraint is expressed algebraically as:

$$\sum_{i=0}^{N-1} x_i = K. \tag{2}$$

Step 3. Following QUBO conventions, the equality constraint is enforced via a quadratic penalty term with weight $A > 0$:

$$\mathcal{P}(x) = A \left(\sum_{i=0}^{N-1} x_i - K \right)^2. \tag{3}$$

Expanding the square yields:

$$\mathcal{P}(x) = A\left(\sum_{i=0}^{N-1} x_i^2 + 2\sum_{0\le i<j\le N-1} x_i x_j - 2K\sum_{i=0}^{N-1} x_i + K^2\right). \tag{4}$$

Applying the idempotent property $x_i^2 = x_i$ for binary variables simplifies the quadratic self-interaction terms:

$$\mathcal{P}(x) = A\left((1-2K)\sum_{i=0}^{N-1} x_i + 2\sum_{0\le i<j\le N-1} x_i x_j + K^2\right). \tag{5}$$

Distributing A gives the general penalty contribution to the Hamiltonian:

$$\mathcal{P}(x) = A(1-2K)\sum_{i=0}^{N-1} x_i + 2A\sum_{0\le i<j\le N-1} x_i x_j + AK^2. \tag{6}$$

Step 4. Combining the negated objective from Step 1 with the expanded penalty from Step 3 yields the full QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{i=0}^{N-1} \left[-w_i + A(1-2K)\right] x_i + \sum_{0\le i<j\le N-1} \left[p_{ij} + 2A\right] x_i x_j + AK^2. \tag{7}$$

Step 5. Reading off the coefficients from the general closed form of $H(x)$, the upper-triangular entries of the QUBO matrix Q and the constant offset c are:

$$Q_{i,i} = -w_i + A(1-2K), \quad \text{for } i \in \{0,\dots,N-1\}, \tag{8}$$

$$Q_{i,j} = p_{ij} + 2A, \quad \text{for } 0 \le i < j \le N-1, \tag{9}$$

$$c = AK^2. \tag{10}$$

The matrix is symmetric, so $Q_{j,i} = Q_{i,j}$ for $i > j$, and all unspecified entries are zero.

Penalty Weight Justification. The penalty weight A must be chosen sufficiently large to ensure that any violation of the exactly-K constraint results in an energy increase that strictly dominates any potential gain in the objective function. The maximum possible reduction in the original objective (or increase in the negated objective) from adding or removing a single sentence is bounded by $\max_i w_i$. Therefore, selecting $A > \max_i w_i$ guarantees that the quadratic penalty term outweighs any feasible objective improvement, forcing the optimizer to prioritize constraint satisfaction over marginal relevance gains.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal summary because the penalty term is strictly non-negative and vanishes if and only if $\sum_i x_i = K$. For any feasible solution satisfying the constraint, $H(x)$ evaluates exactly to the negated objective function. For any infeasible solution, the penalty term $A(\sum_i x_i - K)^2$ adds a positive constant that, by the choice of A, exceeds the maximum possible difference between any two feasible objective values. Consequently, the global minimum of $H(x)$ must occur at a feasible point, and among feasible points, $H(x)$ is minimized exactly when the original objective is maximized.

Illustrative Example. Consider a concrete instance with $N = 3$ candidate sentences and a target summary length of $K = 2$. Let the importance scores be $w_0 = 0.7$, $w_1 = 0.5$, $w_2 = 0.8$, and the redundancy penalties be $p_{01} = 0.1$, $p_{02} = 0.2$, $p_{12} = 0.05$. Setting the penalty weight to $A = 10$ (which satisfies $A > \max_i w_i$), we instantiate the general formulas. The linear coefficients become $Q_{0,0} = -0.7 + 10(1 - 4) = -30.7$, $Q_{1,1} = -0.5 - 30 = -30.5$, and $Q_{2,2} = -0.8 - 30 = -30.8$. The quadratic coefficients become $Q_{0,1} = 0.1 + 20 = 20.1$, $Q_{0,2} = 0.2 + 20 = 20.2$, and $Q_{1,2} = 0.05 + 20 = 20.05$. The constant offset is $c = 10 \cdot 2^2 = 40$. The resulting Hamiltonian is:

$$H(x) = -30.7x_0 - 30.5x_1 - 30.8x_2 + 20.1x_0x_1 + 20.2x_0x_2 + 20.05x_1x_2 + 40. \tag{11}$$

Evaluating $H(x)$ over all binary vectors with exactly two ones yields the following energies: $H(1, 1, 0) = -1.1$, $H(1, 0, 1) = -1.3$, and $H(0, 1, 1) = -1.25$. The global minimum occurs at $x^* = (1, 0, 1)$, corresponding to selecting sentences $\{0, 2\}$. The maximum summary score is $0.7 + 0.8 - 0.2 = 1.3$, confirming that the QUBO correctly identifies the optimal extractive summary.